

SHOALWATER, Australia

WMO: 953700

Lat: 22.470S

Lon: 150.179E

Elev: 92

StdP: 14.65

Time Zone: 10.00 (E10)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	40.7	43.5	29.0	23.1	60.3	33.5	28.2	56.1	16.3	62.7	15.0	63.4	1.0	160	0.326

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.0	90.3	74.7	88.9	74.2	87.6	73.8	79.4	84.1	78.2	83.4	77.2	82.7	9.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	147.7	81.3	76.8	140.2	80.7	75.6	134.7	80.2	43.2	83.8	41.8	83.8	40.9	82.9	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	13.5	12.2	35.6	94.7	2.7	2.5	33.6	96.5	32.0	98.0	30.5	99.4	28.5	101.2
			34.2	81.0	3.6	2.6	31.7	82.9	29.6	84.4	27.6	85.9	25.0	87.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.0	80.4	79.8	78.2	73.7	67.6	63.2	61.9	63.2	68.1	72.7	76.7	79.6	(6)
	DBStd	7.64	2.42	2.41	2.73	2.95	4.21	4.71	4.85	4.49	3.87	3.56	3.04	2.64	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	336	0	0	0	1	25	88	120	88	13	1	0	0	(9)
	CDD50	8040	942	833	874	712	545	396	368	408	544	702	800	916	(10)
	CDD65	2900	477	413	409	263	105	33	23	31	107	238	350	451	(11)
	CDH74	24033	4700	3874	3462	1670	569	120	114	250	760	1562	2726	4226	(12)
	CDH80	6447	1535	1197	954	283	42	2	2	9	79	255	720	1369	(13)
(14) Wind	WSAvg	5.8	6.8	6.5	6.3	5.5	5.0	5.0	4.6	4.8	5.4	6.4	6.6	6.7	(14)
(15) Precipitation	PrecAvg	32.4	6.1	5.8	3.6	2.4	1.5	1.5	1.0	1.1	0.8	1.4	2.2	4.9	(15)
	PrecMax	74.8	15.7	18.2	14.2	8.9	5.7	6.1	6.1	6.4	4.9	3.9	11.2	20.8	(16)
	PrecMin	14.7	1.2	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	11.6	3.9	4.8	3.7	2.5	1.3	1.3	1.5	1.5	1.1	1.1	2.5	4.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.8	91.8	90.8	86.6	83.3	79.4	78.7	81.0	85.4	86.8	90.2	92.1	(19)
		MCWB	76.1	76.1	75.1	72.0	68.7	66.0	64.5	65.0	67.1	70.0	71.8	72.1	(20)
	2%	DB	89.7	89.9	88.2	84.5	81.0	77.0	76.8	78.5	81.9	84.2	87.1	89.3	(21)
		MCWB	75.3	75.5	74.7	71.3	67.5	64.7	63.8	64.2	66.9	69.9	71.8	73.2	(22)
	5%	DB	88.2	88.2	86.6	83.1	79.3	75.2	75.1	76.7	80.2	82.7	85.5	87.8	(23)
		MCWB	74.6	75.0	74.1	70.8	66.9	64.0	63.2	63.5	66.0	69.0	71.1	72.8	(24)
	10%	DB	86.8	86.4	85.0	81.6	77.3	73.5	73.2	74.9	78.4	81.3	84.2	86.4	(25)
		MCWB	74.1	74.7	73.5	70.4	66.1	63.4	62.5	62.6	65.5	68.1	70.6	72.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.6	81.7	79.8	79.1	74.3	70.2	70.8	70.1	72.8	74.3	78.5	78.6	(27)
		MCDB	85.7	85.4	84.2	82.1	78.0	74.5	72.9	75.9	79.5	81.2	82.6	85.5	(28)
	2%	WB	79.1	79.7	78.1	76.9	71.4	68.4	68.3	68.8	71.4	73.0	76.0	77.4	(29)
		MCDB	84.0	84.2	83.0	80.4	76.2	72.8	73.3	74.1	76.7	79.1	81.5	83.6	(30)
	5%	WB	77.9	78.0	76.9	74.8	69.7	67.0	66.8	67.0	70.0	72.0	74.2	76.5	(31)
		MCDB	83.3	83.5	82.2	78.8	75.0	71.8	71.7	73.3	75.9	78.9	80.9	82.6	(32)
	10%	WB	76.7	76.9	75.8	72.9	68.3	65.6	65.0	65.1	68.4	70.7	73.0	75.5	(33)
		MCDB	82.8	82.5	81.5	77.8	74.2	71.0	70.0	72.1	74.9	77.9	80.5	81.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	12.8	13.9	16.4	20.0	20.7	23.2	24.5	21.6	17.9	15.3	13.7	(35)
		MCDBR	15.3	16.0	16.1	18.4	21.8	22.5	24.1	25.2	23.0	19.3	17.0	15.3	(36)
	5% WB	MCWBR	4.3	4.8	5.1	7.7	9.7	11.5	12.6	12.7	9.9	7.1	5.4	4.3	(37)
		MCWBR	12.5	12.6	13.3	14.5	16.3	16.3	16.6	19.5	16.8	15.7	14.1	12.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.398	0.390	0.374	0.352	0.331	0.321	0.311	0.324	0.351	0.377	0.391	0.394	(40)
		taud	2.510	2.538	2.570	2.610	2.623	2.647	2.651	2.596	2.513	2.472	2.467	2.492	(41)
	Ebn at Noon		300	298	295	289	283	280	287	294	298	299	300	302	(42)
		Edh at Noon	36	35	32	29	26	24	25	29	34	36	38	37	(43)
(44) All-Sky Solar Radiation	RadAvg		2042	1899	1738	1507	1301	1139	1289	1565	1900	2100	2171	2159	(44)
	RadStd		137	204	145	138	71	105	94	105	131	126	231	215	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page